Finite Field Extensions. Existence and Practical Usage

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Dmytro Zakharov
Kyiv School of Economics
Mathematics Department

KSE | Kyiv School of Economics

Introduction

Prime Fields

Definition (Field)

Recall that $\mathbb{K}$ is a **field** if:

- $(\mathbb{K}, +)$ and $(\mathbb{K} \setminus \{0\}, \times)$ form abelian groups.
- Distributive law holds: $a(b+c) = ab + ac$ for arbitrary $a, b, c \in \mathbb{K}$.

Field $\mathbb{K}$ is **finite** if it contains finite number of elements.

The most basic example of such field is the **prime field** $\mathbb{Z}/p\mathbb{Z}$, further denoted simply as $\mathbb{F}_p$. For instance, for $p = 5$ one has:

$$2 + 4 = 1, \quad 3 \cdot 4 = 2, \quad 3^{-1} = 2 \quad (\textit{all over } \mathbb{F}_5)$$

Motivating Question

However, is this the only form of finite field?

Constructing $\mathbb{F}_{p^2}$

Let us give some example of the field with p^2 elements.

Example

Suppose it so happens that $p \equiv 3 \pmod 4$. Then, the following quotient ring

$$\mathbb{F}_{p^2} \cong \mathbb{F}_p[X]/\langle X^2 + 1 \rangle$$

is a finite field with p^2 elements.

Proof. Let us carefully go step-by-step:

1. $\mathbb{K}[X]/\langle f(X) \rangle$ is a field iff $f(X) \in \mathbb{K}[X]$ is irreducible over $\mathbb{K}$.
2. Thus, we prove that $X^2 + 1$ is irreducible over such $\mathbb{F}_p$.
3. We note that -1 is a square in $\mathbb{F}_p$ iff $p \equiv 1 \pmod 4$. Thus, -1 is not a square when $p \equiv 3 \pmod 4$, so $X^2 + 1$ is irreducible.
4. Thus $\mathbb{F}_p[X]/\langle X^2 + 1 \rangle$ is a field. It obviously contains p^2 elements since there are p^2 linear polynomials over $\mathbb{F}_p$.

More on $\mathbb{F}_{p^2}$

Remark #1. Such construction is typically denoted as $\mathbb{F}_p(i)$ where $i^2 = -1$. We will see why.

Remark #2. Moreover, $\mathbb{F}_{p^2}$ is a *field extension* of $\mathbb{F}_p$.

Definition (Field Extension)

Let $\mathbb{F}$ be a field. $\mathbb{K}$ is called a **field extension** if (i) $\mathbb{K}$ is a field, (ii) $\mathbb{F} \subseteq \mathbb{K}$. This is commonly denoted as $\mathbb{K}/\mathbb{F}$.

Obviously, all elements in $\mathbb{F}_p$ are contained in $\mathbb{F}_{p^2}$, so we have an extension $\mathbb{F}_{p^2}/\mathbb{F}_p$. Let us give an example.

Example

Pick $p := 19$. Then, $\mathbb{F}_{361} \cong \mathbb{F}_{19}(i)$. Example of operations:

$$\textbf{Addition:} \quad (1 + 10i) + (18 + 15i) = 19 + 25i = 6i$$

$$\textbf{Multiplication:} \quad (5 + 6i)(6 + 7i) = 30 + 71i + 42i^2 = 7 + 14i.$$

Extending to $\mathbb{F}_{p^n}$

In general, for $\mathbb{F}_p$, we can always find irreducible $f(X) \in \mathbb{F}_p$ with $\deg f = n$, so $\mathbb{F}_{p^n}$ is constructed as:

$$\mathbb{F}_{p^n} \cong \mathbb{F}_p[X]/\langle f(X) \rangle.$$

Today's Goal: Why this is true?

Example (Binary Field Extensions)

Once, I was asked, why the library *Binius*, operating over binary field extensions, gives

$$3 + 5 = 6$$

Turns out this is done over $\mathbb{F}_{2^8} \cong \mathbb{F}_2[X]/\langle X^8 + X^4 + X^3 + X + 1 \rangle$. So that said, we have:

$$3 \leftrightarrow 1 + X,\ 5 \leftrightarrow 1 + X^2 \Rightarrow 3 + 5 \leftrightarrow X^2 + X \leftrightarrow 6.$$

More on Field Extensions

Example (Cont. on $\mathbb{F}_{2^8}$)

Moreover, we can have more ridiculous statements:

$$128 \times 15 = 193 \text{ (over } \mathbb{F}_2[X]/\langle X^8 + X^4 + X^3 + X + 1\rangle).$$

Indeed, $128 \leftrightarrow X^7$, $15 \leftrightarrow X^3 + X^2 + X + 1$. One then checks:

$$X^7(X^3+X^2+X+1) \equiv X^7+X^6+1 \pmod{X^8+X^4+X^3+X+1}$$

However, is the only way to construct $\mathbb{F}_{p^n}$ is to consider quotient $\mathbb{F}_p[X]/\langle f(X)\rangle$ with $\deg f = n$? Let $m \mid n$. Then,

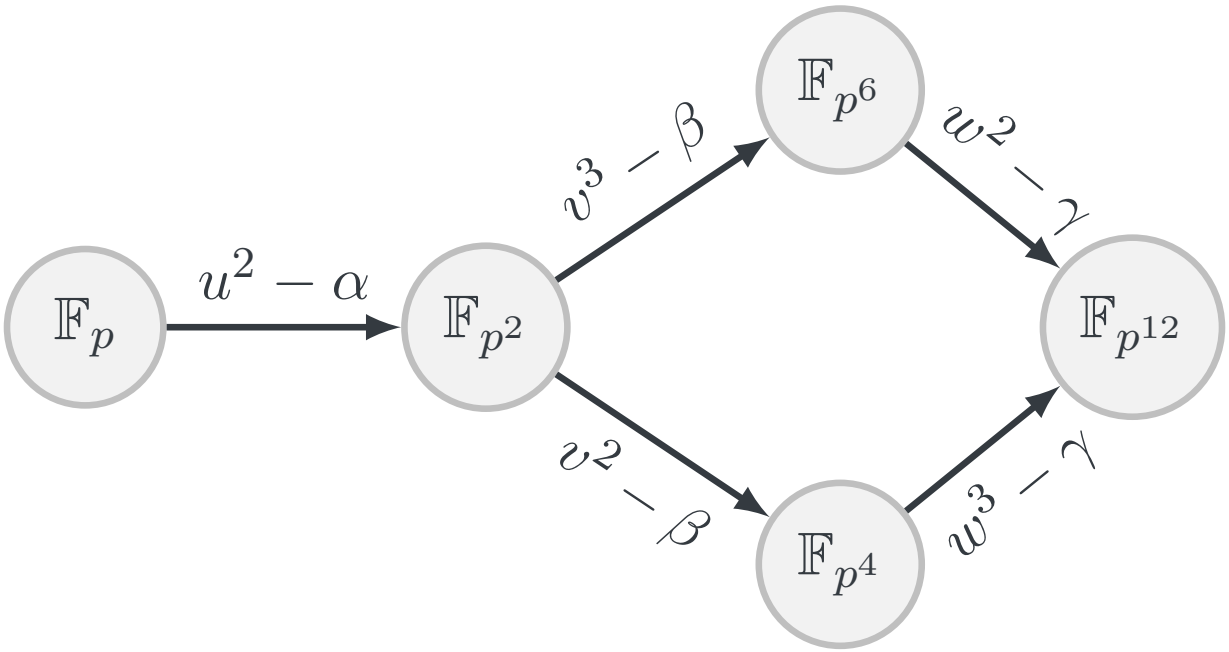
$$\mathbb{F}_{p^n} \cong \mathbb{F}_{p^m}[X]/\langle g(X)\rangle, \quad g(X) \in \mathbb{F}_{p^m}[X],\ \deg g = n/m.$$

So we have a field extension $\mathbb{F}_{p^n}/\mathbb{F}_{p^m}$ for $m \mid n$. Again, it is unobvious why such construction always exists.

Example: BN254 Field Extensions

The last fact is commonly exploited in cryptographic protocols implementation. For instance, in BN254, one needs to perform arithmetic over $\mathbb{F}_{p^{12}}$ with:

$$p = \texttt{0x30644E7}\ldots\texttt{C208C16D87CFD47}$$



Turns out, that such tower of extensions give (i) for upper path, better speed, (2) for lower path, better compression. See [Hou23].

Tower of Fields

Example (BN254 Tower of Fields)

One has the following construction:

$$\mathbb{F}_{p^2} \cong \mathbb{F}_p[u]/\langle u^2 + 1\rangle,$$
$$\mathbb{F}_{p^6} \cong \mathbb{F}_{p^2}[v]/\langle v^3 - \xi\rangle,$$
$$\mathbb{F}_{p^{12}} \cong \mathbb{F}_{p^6}[w]/\langle w^2 - v\rangle,$$

with $\xi = 9 + u$.

Existence

Basic properties of Finite Fields

Further, assume that $\mathbb{K}$ is an aribtrary finite field of size q.

Lemma

$$X^q - X = \prod_{\alpha \in \mathbb{K}} (X - \alpha)$$

Proof. Every $\alpha \in \mathbb{K}$ is a root in both sides of equation. Polynomials are of same degree, which proves the equality.

Corollary 1. Let $\mathbb{K}'/\mathbb{K}$ be the field extension. Then, an element $\alpha \in \mathbb{K}'$ is in $\mathbb{K}$ iff $\alpha^q = \alpha$.

Corollary 2. If $f(X) \mid (X^q - X)$, then $f(X)$ has $\deg f$ roots.

Lemma

The integer multiples of the identity form a subfield of $\mathbb{K}$ isomorphic to $\mathbb{Z}/p\mathbb{Z}$ for some prime p.

Why $\#\mathbb{K} = p^n$?

Proof. Let e be the multiplicative identity of $\mathbb{K}$. Define morphism $\phi : \mathbb{Z} \to \mathbb{K}$ as $n \mapsto ne$. It is clearly a ring homomorphism. Moreover, $\operatorname{im}\phi$ is a finite subring of $\mathbb{K}$, so it is an integral domain; $\ker\phi$ is a non-zero prime ideal. Thus, $\operatorname{im}\phi \cong \mathbb{Z}/p\mathbb{Z}$ for prime p.

This way, we shall think of $\mathbb{K}$ as a finite dimensional space over $\mathbb{Z}/p\mathbb{Z}$. Let n be such dimension and let $\omega_1, \ldots, \omega_n$ be its basis. Then every element $\omega \in \mathbb{K}$ and be expressed uniquely as $a_1\omega_1 + \cdots + a_n\omega_n$ with $a_i \in \mathbb{Z}/p\mathbb{Z}$. It follows that $\mathbb{K}$ has p^n elements. Thus, we have proven

Lemma

The number of elements in the finite field is a power of prime.

We call this prime a **characteristic** of $\mathbb{K}$.

Other Basic Facts

Lemma

Suppose $p = \mathrm{char}(\mathbb{K})$. *Then, for any* $\alpha, \beta \in \mathbb{K}$ *one has* $(\alpha + \beta)^{p^d} = \alpha^{p^d} + \beta^{p^d}$ *for any integer* $d > 0$.

Multiplicative Group $\mathbb{K}^\times$

Theorem

*Any multiplicative group $\mathbb{F}_q^\times$ is **cyclic** with $q-1$ elements.*

Proof.

References I

[Hou23] Youssef El Housni. “Pairings in Rank-1 Constraint Systems”. In: *Applied Cryptography and Network Security*. Ed. by Mehdi Tibouchi and XiaoFeng Wang. Cham: Springer Nature Switzerland, 2023, pp. 339–362. ISBN: 978-3-031-33488-7.

Any Questions?

